# Matrices 1 Assessment

1) Which of the following matrices is an identity matrix?

Circle your answer.

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \qquad \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

**[1 mark]**

2) Which of the following expressions is the determinant of the matrix $\begin{bmatrix} a & 2 \\ b & 5 \end{bmatrix}$ ?

Circle your answer.

$5a - 2b \qquad 2a - 5b \qquad 5b - 2a \qquad 2b - 5a$

**[1 mark]**

3) $\mathbf{A} = \begin{bmatrix} 2 & 3 \\ k & 1 \end{bmatrix}$

a) Find $\mathbf{A}^{-1}$

**[2 marks]**

b) The determinant of $\mathbf{A}^2$ is equal to 4. Find the possible values of $k$.

**[3 marks]**

4) Two matrices $\mathbf{A}$ and $\mathbf{B}$ satisfy the equation

$$\mathbf{AB} = \mathbf{I} + 2\mathbf{A}$$

where $\mathbf{I}$ is the identity matrix and $\mathbf{B} = \begin{bmatrix} 3 & -2 \\ -4 & 8 \end{bmatrix}$

Find $\mathbf{A}$.

**[3 marks]**

5) $\mathbf{A}$ and $\mathbf{B}$ are non-singular square matrices.

(a) Write down the product $\mathbf{AA}^{-1}$ as a single matrix. **[1 mark]**

(b) $\mathbf{M}$ is a matrix such that $\mathbf{M} = \mathbf{AB}$.

Prove that $\mathbf{M}^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$.

**[3 marks]**

6) (a) Show that the matrix $\begin{bmatrix} 5-k & 2 \\ k^3+1 & k \end{bmatrix}$ is singular when $k = 1$ **[1 mark]**

(b) Find the values of $k$ for which the matrix $\begin{bmatrix} 5-k & 2 \\ k^3+1 & k \end{bmatrix}$ has a negative determinant.

Fully justify your answer

**[5 marks]**

7) The matrix **A** is given by $\mathbf{A} = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$

(a) Given that $\mathbf{A}^2 = \begin{bmatrix} p & -2 & -4 \\ 5 & 6 & 4 \\ 10 & q & 9 \end{bmatrix}$, find the value of $p$ and the value of $q$.

**[2 marks]**

(b) Given that $\mathbf{A}^3 - 6\mathbf{A}^2 + 11\mathbf{A} - 6\mathbf{I} = \mathbf{0}$, prove that

$$\mathbf{A}^{-1} = \frac{1}{6}(\mathbf{A}^2 - 6\mathbf{A} + 11\mathbf{I})$$

**[2 marks]**

(c) Given that $\mathbf{A}^{-1} = \frac{1}{6}\begin{bmatrix} r & -2 & -2 \\ -1 & 5 & -2 \\ -2 & s & 2 \end{bmatrix}$, find the value of $r$ and the value of $s$.

**[2 marks]**

(d) Hence, or otherwise, find the solution of the system of equations

$$\begin{aligned} x \qquad - z &= k \\ x + 2y + z &= 5 \\ 2x + 2y + 3z &= 7 \end{aligned}$$

giving your answers in terms of $k$.

**[3 marks]**

**Total: 29 marks**